

Driver related

1 About Python

Q: What do the parameters in the `send_coords([x,y,z,rx,ry,rz], speed, 1)` API mean? What do `rx`, `ry`, and `rz` correspond to in Euler angles? What is the rotation order of Euler angles? And what is the range of values for each parameter?

- A: The parameters in the previous array are the end coordinates of the robot, speed is the speed, and the last parameter is the motion mode. `rx`, `ry`, and `rz` should correspond to `rx`, `ry`, and `rz`, which corresponds to roll, pitch, and yaw respectively. The order of Euler angles is `zyx`, and `zyx` is the local coordinate. The value range of `X`, `Y`, and `Z` is `-300~300.00` (the value range is undefined, and the inverse kinematics no solution prompt will be returned if it exceeds the range), and the value range of `rx`, `ry`, and `rz` is `-180~180`.

Q: Are sample tutorials for the Python API provided?

- A: There is a demo available. There is a test code in the Test folder of github. You can execute it with the terminal. " https://github.com/elephantrobotics/pymycobot/tree/main/demo/myCobot_320_demo "

Q: Do different versions of the robotic arm use the same Python API?

- A: The API is the same.

2 About ROS

Q: How do microcontroller-based myCobot and microprocessor-based myCobot run ROS?

- A: The microcontroller-based robotic arm is currently on Ubuntu. You need to download a virtual machine to build a virtual environment and install the Ubuntu system in the virtual environment to run ROS. You can also develop your own ROS. The microprocessor-based robotic arm comes with a ROS environment and can be used directly.

Can a microprocessor based robotic arm be connected to a pc to use ros and moveit?

- A: The current open source data does not have direct communication control. It can be achieved by modifying the existing node files through `ros + socket`.

Q: Can you provide rviz model files and programming examples?

- A: It is available on our github. " https://github.com/elephantrobotics/mycobot_ros "

Q: When using ROS to start the rviz model file, why does it report an error saying permission denied: '/dev/ttyUSB0'?

- A: It is because the serial port permissions are not given. You should enter `sudo chmod 777 /dev/ttyUSB0` port name in the terminal. For example:

```
sudo chmod 777 /dev/ttyUSB0
```

Q: When running the slider control and model following commands of ROS, why does the error `init()` take exactly 2 arguments (3 given) appear?

- A: The `pymycobot` library is not installed and started.

Q: When using ROS, why are the robot arm angles inconsistent with the model angles after opening the rviz model?

- A: It is very likely that the zero position of the robot arm is not calibrated. The zero position of the robot arm needs to be calibrated.